

Social Media Bot Detection

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INTRODUCTION

Problem Statement

Over 50% of internet activity is bot-generated, with malicious bots being responsible for largescale fraud, misinformation, and resource exhaustion on social media platforms. Despite the existence of current bot detection systems, most of these solutions rely on surface-level indicators such as IP addresses, traffic volume, and content.

Background

In September 2025, the Secret Service discovered and eliminated a network of more than 300 SIM servers and 100,000 SIM cards in an attempt to conduct multiple telecommunication attacks against U.S. government officials. [1] Which is why we focused on SIM farms for our simulation.

Design and Implementation

The architecture below shows how the different users; bots and humans connect to the server. Human users "bounce" between Wi-fi and cellular (5G) whereas bot users solely connect to Wi-Fi.

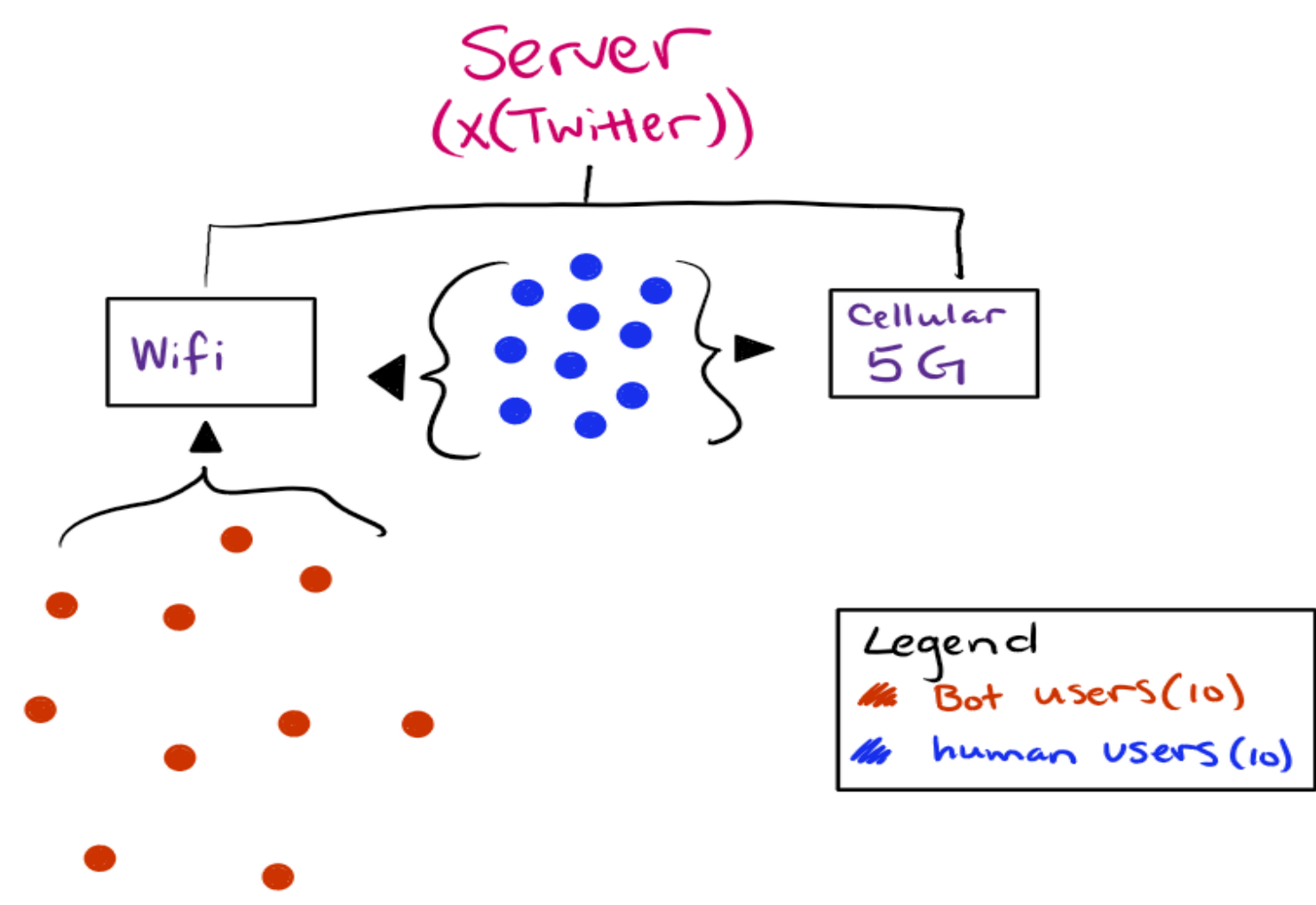


Figure 1: Network Architecture. The overall architecture was designed to replicate multiple bots and users making requests to Twitter server. We include a central TCP server that receives all traffic, the 5G network(gNB+ EPC (PGW, SGW, MME)), and 802.11n AP for the Wi-Fi connectivity.

We built our simulation using ns-3 and real Twitter data from a CSV containing usernames, tweet text, timestamps, and bot/human labels, enabling realistic network traffic simulation.

RESULTS

From our simulation, we were able to trace several key performance indicators (KPIs) of the network such as packets sent/received, packet loss, delay, throughput, etc. We examined several patterns shown in the figures below. Across all metrics, bots consistently display low variance with few outliers, whereas humans show wide distributions. These patterns can provide robust signatures for bot detection systems and reveal behavioral differences that distinguish automated traffic from human activity.

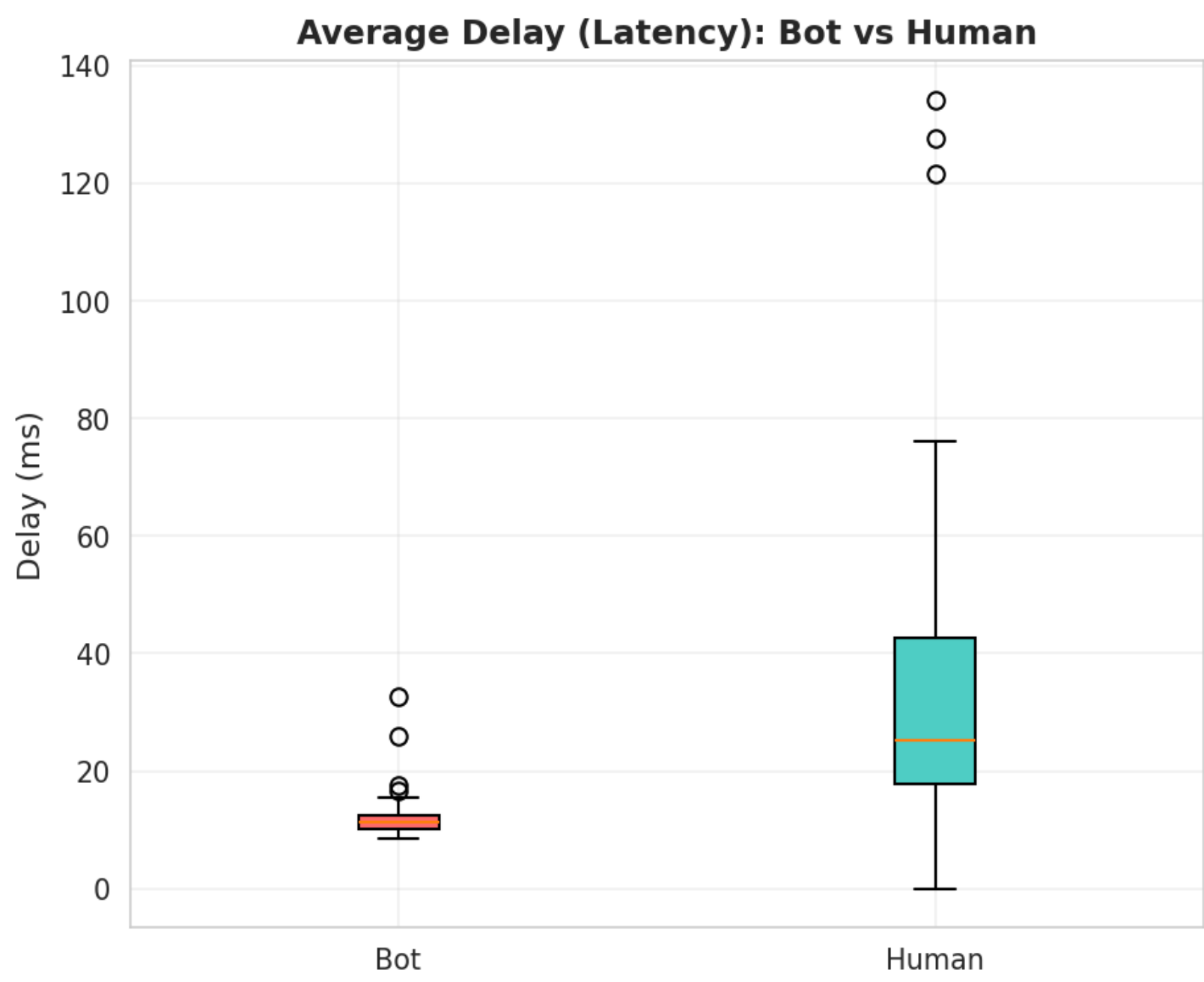


Figure 2: Average Latency Comparison. Bots maintain a consistent low latency whereas human traffic experiences much higher and more variable delays, due to varied user activities and movement.

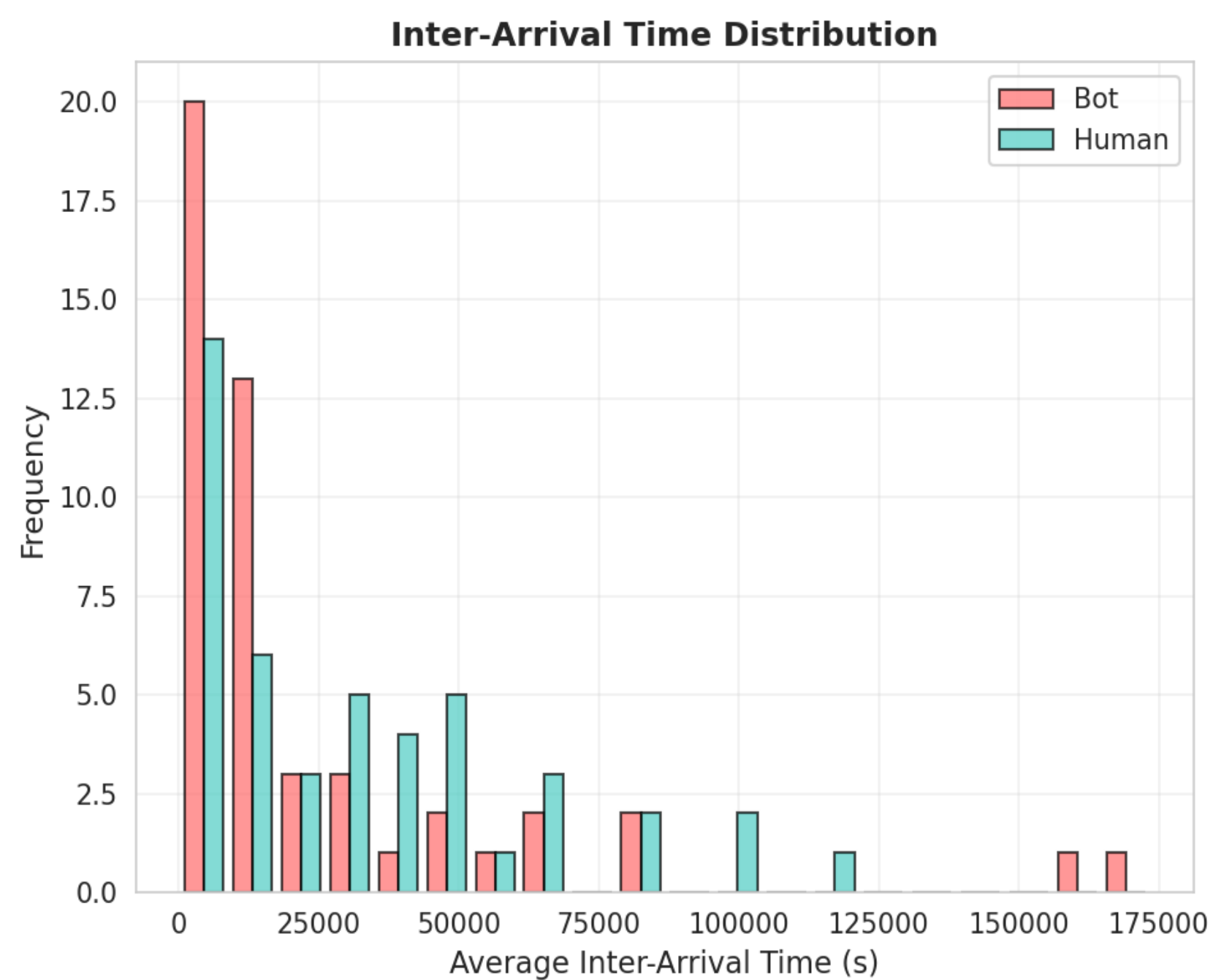


Figure 5: Inter-Arrival Time Distribution Comparison. The inter-arrival time is the time in between each consecutive network packet (the time between tweets). Bots exhibit a highly concentrated distribution with most inter-arrival times clustered in the 0-25,000 second range, suggesting that bots generate traffic in rapid and regular bursts. Humans show a more gradual dispersed distribution suggesting that humans have both periods of active engagement and irregular breaks.

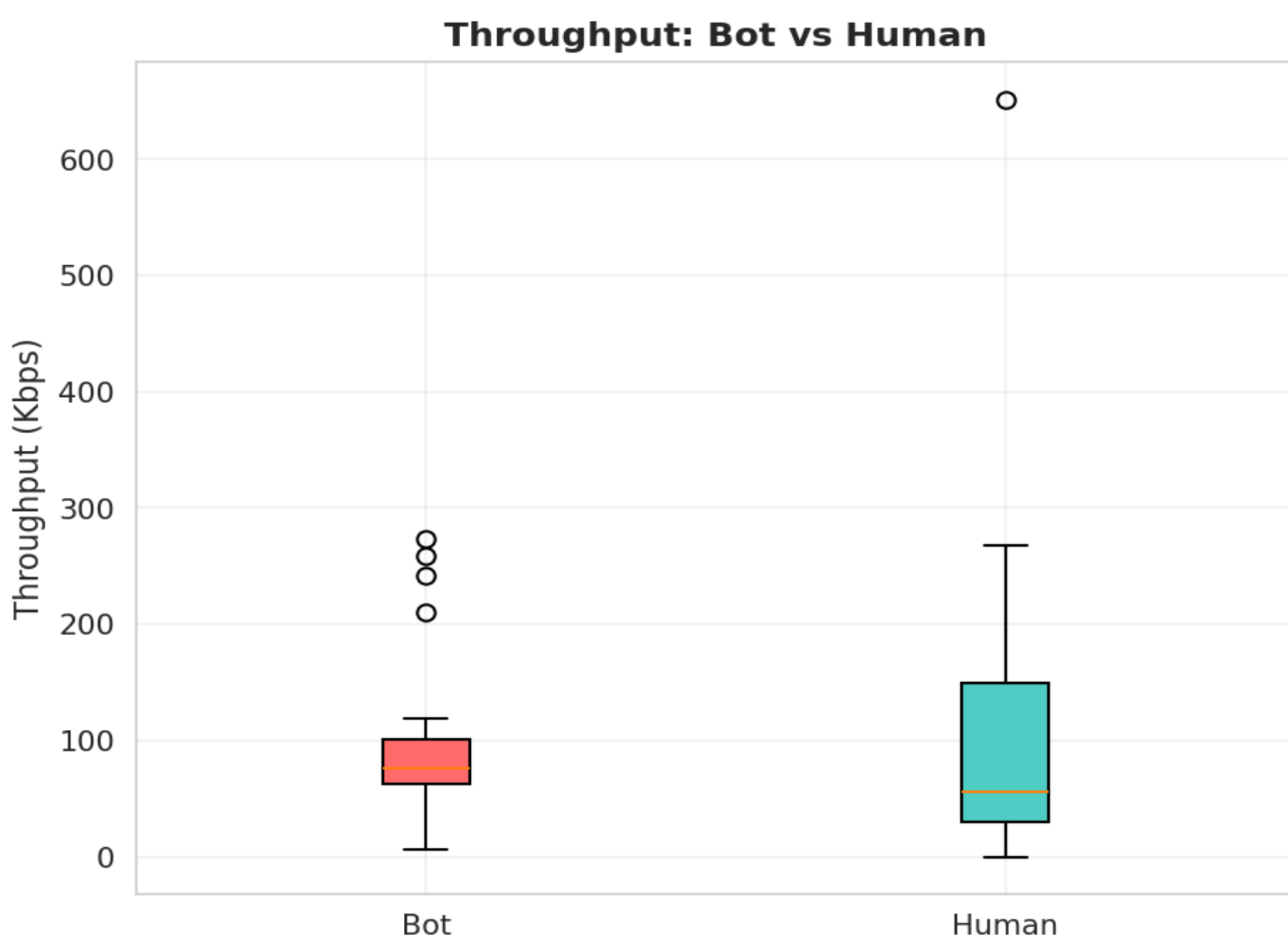


Figure 3: Throughput comparison. Bots and humans both have significant throughput variability. Bots have a slightly higher median throughput indicating that bots are have more bursts of activity.

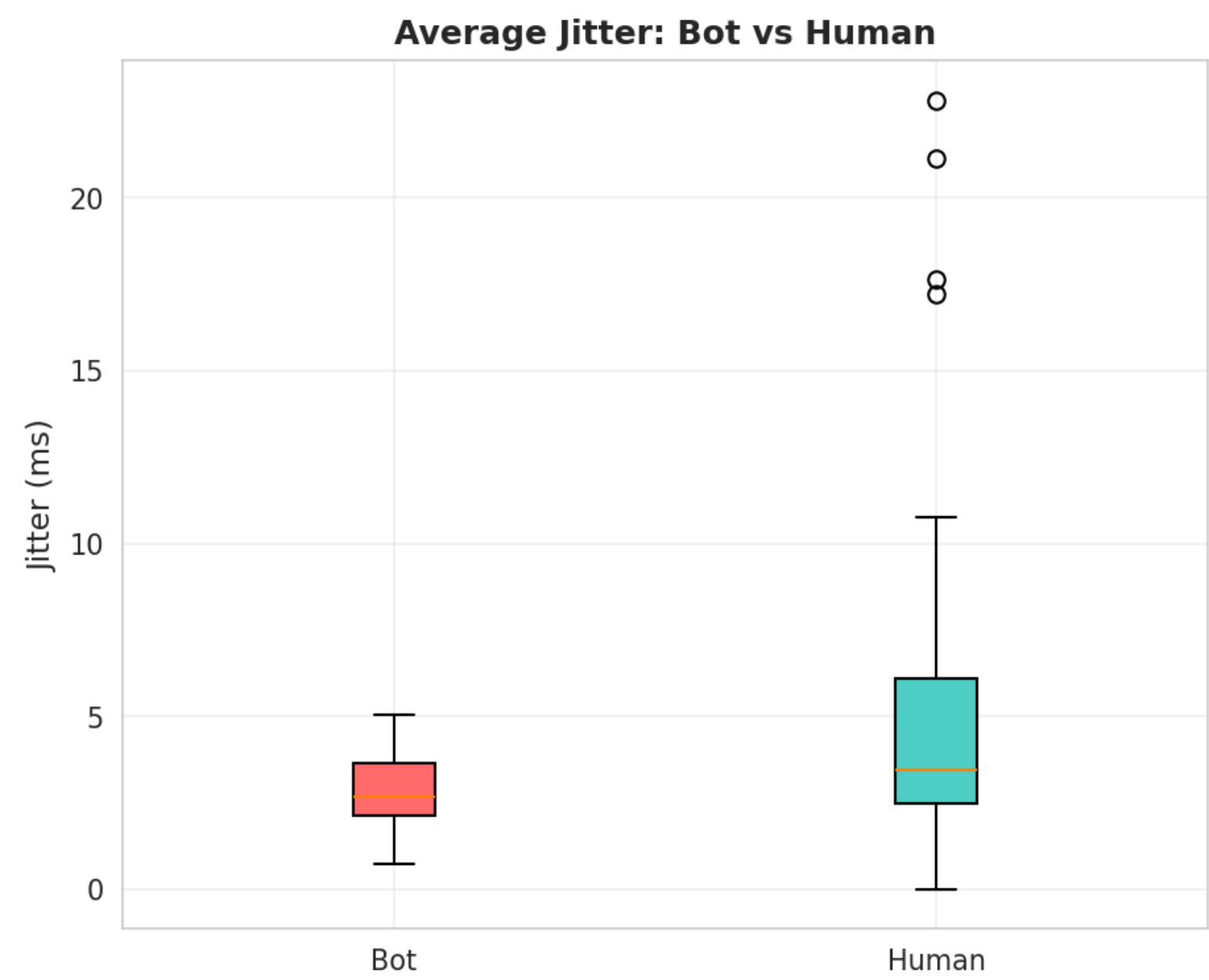


Figure 4: Average Jitter Comparison. Bots maintain a consistent low jitter whereas human traffic demonstrates a much higher and variable jitter reflecting the unpredictable nature of the human behavior on networks.

RESULTS (CONT.)

The UI takes in a csv file as an input. It then runs the networking simulation with that data file, generating a variety of outputs. Those results are then formatted and displayed as downloadable files, viewable tables, and network animations.

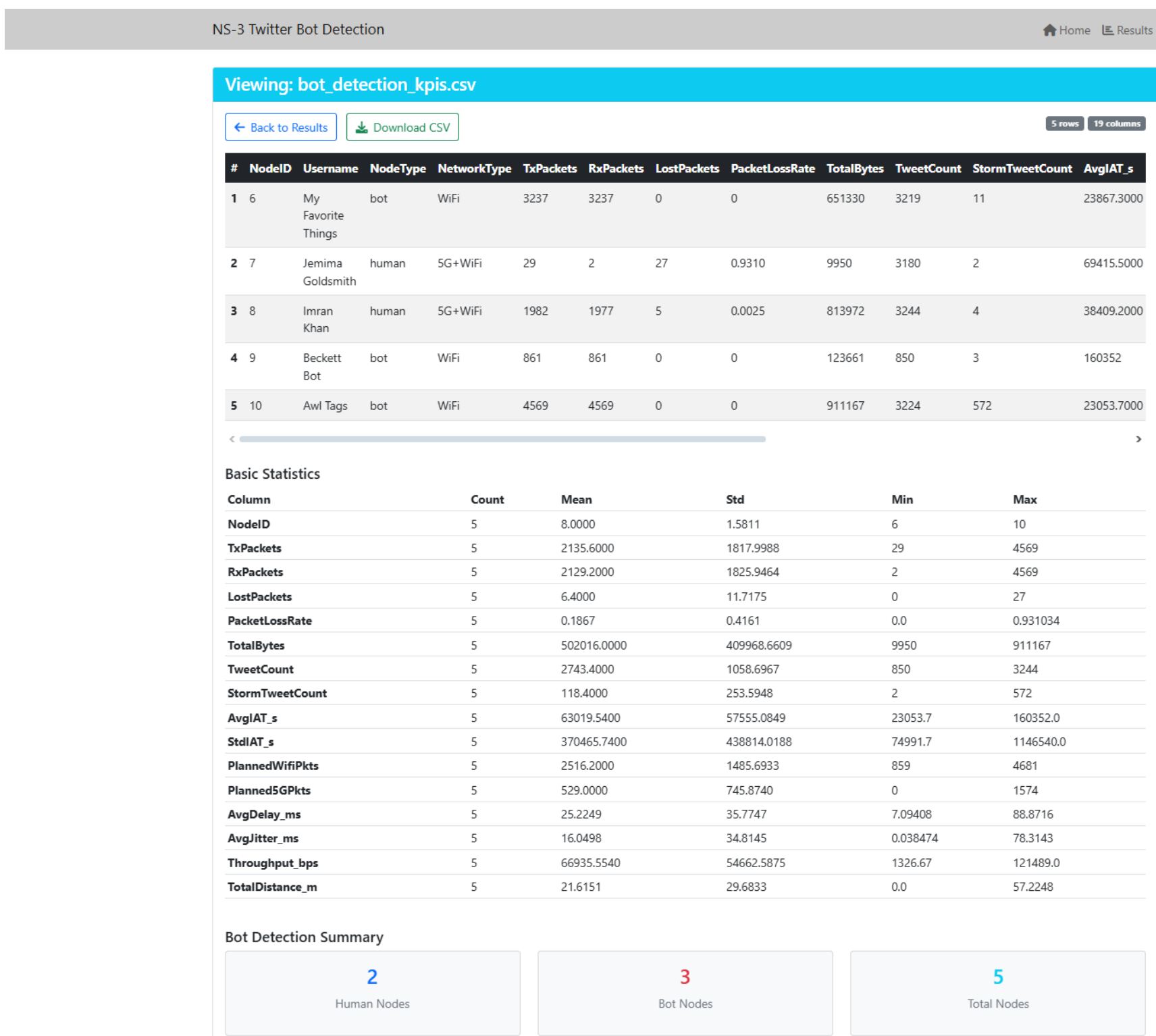


Figure 6: User Interface Results. The User Interface aspect offers a user-friendly way to utilize the Network simulation and visualize the results

CONCLUSION

- Our simulation demonstrates that bots and humans produce fundamentally different network behaviors, especially in latency, jitter, throughput, and inter-arrival times. These differences provide a strong foundation for developing network-level bot detection features beyond traditional methods.
- We learned that network-level KPIs naturally separate bots from humans, and that behavioral patterns are strong indicators of user type.
- Future work includes using the collected KPIs to create a machine learning algorithm capable of classifying human and bot users.

REFERENCES & ACKNOWLEDGEMENTS

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[1] <https://www.secretservice.gov/newsroom/releases/2025/09/us-secret-service-dismantles-imminent-telecommunications-threat-new-york>